The Living Economy

A complete design for the economy of Rise of Mankind: land, people, goods, living standards, industry, markets, trade, the state and war, as one system

This document describes the economy as it should be, whole, rather than a sequence of patches. It replaces the map's resource tokens, the on/off resources, the leasing of resources for gold, and the fixed yields of specialists with one ledger of goods. That ledger is made by people working land and workshops, consumed by the population according to its standard of living, transformed by industry, priced by markets, carried by trade and taxed by the state. The game's three yields and four commerce outputs remain what the player sees, because the executable's screens show exactly those; underneath, they become views of the ledger.

What it rests on. Every fact about today's game is from the mod's XML and DLL at GitLab `main 3bec049`. Every measured number is from *Land, not tokens* and *Settlement Economy* (research/economy): 30 real harness games, rebuilt city by city, and a deterministic model of goods run on them. Every parameter is either an anchor taken from the engine, a quantity derived from those measurements, or a design intent stated as such (section 17). The need levels in this document were not chosen: they were solved for, so that the stated targets are met in the 30 games (section 17.3).

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1. Playing it

2000 BC. Your capital sits where a river crosses warm plains. The overlay shows why the site was good: wheat suits the flood plain at 0.8, the hills to the north hold copper of grade 0.7, the coast is shelf water. The city of size 4 has two farmers, one herder and one fisher; it eats 12 units of staple a turn and grows from the surplus as it always did. Its people also want a little meat, wool and firewood; the ledger shows wool short, so a weaver's slot sits empty and two citizens wear rags, which costs one point of contentment. A neighbour offers wool for your copper. You accept a contract: 2 wool a turn for 1 copper a turn, 20 turns.

400 AD. Eleven cities, one network of roads and a river. The market shows grain cheap and wine dear; your coastal cities are fed by the inland farms, which is why the iron town in the hills could grow to 9 on rock. You know olives and grapes, not rice or sugar; a merchant from the south sells you rice seed for 300 gold, and the river delta that was marsh becomes paddy over the next ten turns. Your iron province is running thin at the shallow layer; Explosives is three techs away. You build an arsenal and start putting by arms, because the empire to the east has twice your swords and its supply of tin runs through your port.

1850. Coal and iron meet in the north; four mills turn them into steel, and the steel into rails and rifles. A third of your people now work in workshops and factories rather than fields, fed by grain bought abroad under a 50-turn contract you renegotiate every war. The cotton you import is turned into cloth and sold back at three times its price. Your rival blockades the strait; the price of cotton doubles in a turn, two mills stand idle, and contentment in the textile towns falls. You can pay, fight, or plant cotton in the south, which the climate overlay says will grow at 0.4.

Every one of these situations follows from the rules below. None needs a scripted event.

2. Pillars

Pillar	Meaning	Consequence for the design
Material truth	Things are made of things. Nothing appears from nowhere or is used without being made	One conserved ledger; recipes consume inputs; armies cost goods
The land matters	Where you are decides what you can do easily, and what you must trade for	No tokens; climate, soil, ore and sea as qualities of every plot
People are the economy	Output is work; what people need decides what they do and how content they are	Occupations; living standards by tier; needs grow with size
Legibility	Every number can be traced to its cause in two clicks	Ledger screen; overlays; help text that names the plot, worker and building
Consequence without micromanagement	Choices are about direction, not clicks	Automatic distribution within networks; governors that know needs; contracts that run themselves
AI parity	The AI plays the same economy and understands it	One valuation engine for every AI decision (section 15)
Determinism	Every computer reaches the same result, bit for bit	Integers in hundredths; exact rationing; the game's synchronised random numbers

3. The whole system

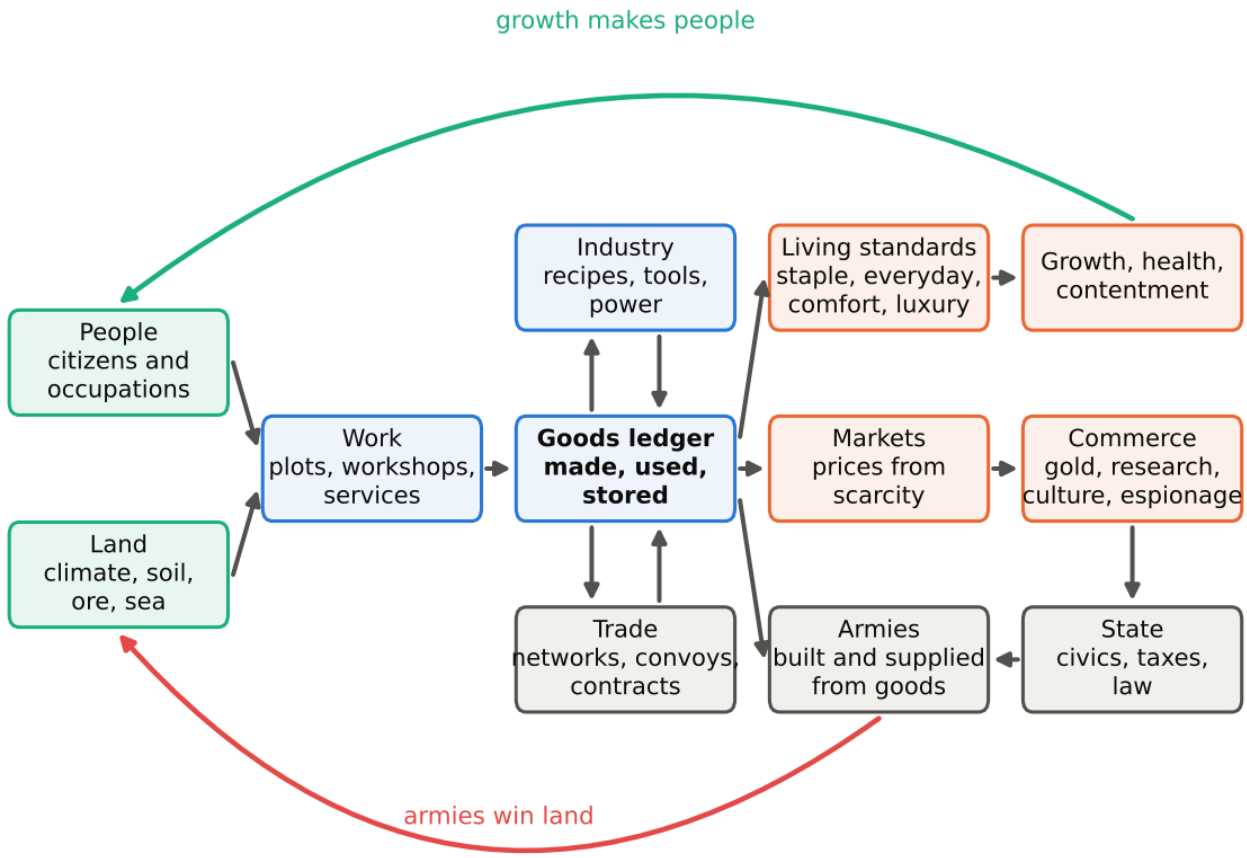


Figure 1. The economy as one system. Land and people produce work; work fills the goods ledger; goods feed living standards, industry, markets, trade and armies; markets produce commerce for the state; growth makes people; armies win land; policy shapes work.

There are three loops, and every mechanic belongs to at least one. The **loop of life**: land and work make staple and everyday goods, which make people grow and stay content, which makes more work. The **loop of wealth**: surplus goods are sold in markets, commerce is taxed into gold, research and culture, which buy buildings and technology that make work more productive. The **loop of power**: goods make and supply armies, armies take or protect land and trade, and land and trade feed the other two loops.

The design's central measured fact is in figure 2: in the 30 games, the worked plots account for nearly all production at first and for about a quarter of it by turn 150 (production 4.08 times what the plots give, middle half 3.1 to 5.1). A game of Rise of Mankind is the passage from an economy of land to an economy of capital. The Living Economy makes that passage visible and material: the same share that is a modifier today becomes workshops, tools, power and the goods they consume.

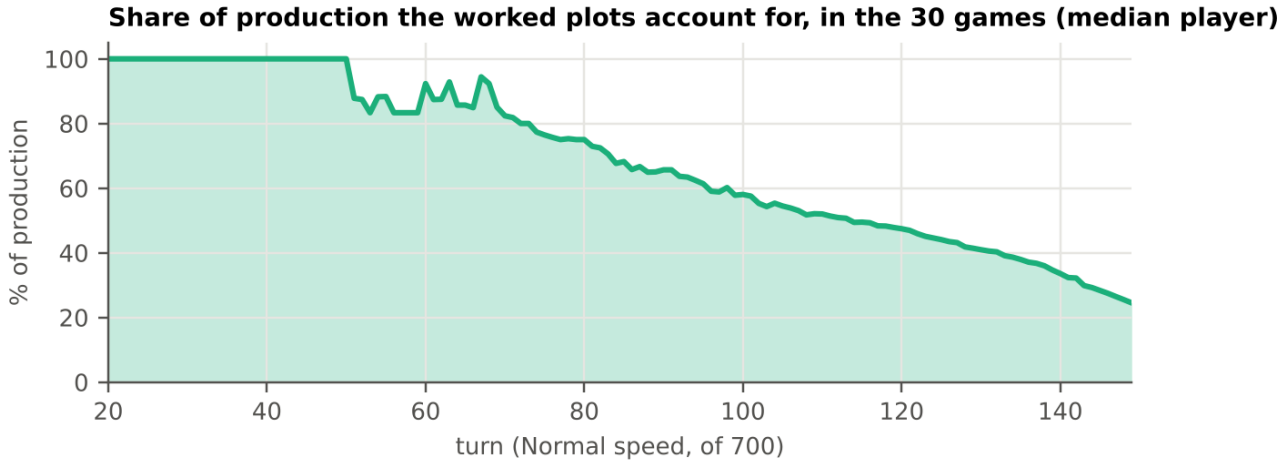


Figure 2. The share of production that the worked plots account for, as the median player's plots' production divided by its observed production (CvCity::getYieldRate), in the 30 games. The rest comes from buildings, civics and specialists.

4. What is cut, replaced and kept

Today	In the Living Economy	Why
Resource tokens on plots; improvements that unlock them	Cut. Land qualities: crop suitability, ore grade and depth, fishing grounds	Places get character; resources follow geography (Land, not tokens)
A resource is on or off for a network	Cut. Goods are quantities, made and used each turn	Scarcity, specialisation and trade become possible
+1 happiness or health per resource, whatever the city's size	Replaced. Living standards by tier, met per citizen	Large cities need more; luxury becomes a matter of supply
Leasing a resource copy for gold per turn	Replaced. Shipments, contracts, seeds and livestock, packages	Between AIs, 568 of 580 resource deals were swaps of copies and 1 paid gold
Buildings that need resources present and give a copy	Replaced. Recipes that turn inputs into outputs, run by workers	Factories should use what they need
Specialists with fixed yields	Replaced. Occupations that produce goods (smith, weaver, miller) or services (merchant, scholar, priest)	Work is what makes goods
Trade routes worth 1.00 commerce times modifiers	Replaced. Convoys that carry goods, earning on what they carry	On this map 2708 of 3025 routes paid exactly the floor
Food, production, commerce on the city screen	Kept, as views of the ledger	The executable's screens show exactly these
Gold, research, culture, espionage sliders	Kept. They split taxed commerce as today	Same
Growth thresholds, food per citizen	Kept: 3 staple per citizen, threshold 30 + 10 × size	The pacing of 700 turns rests on them
Civics, technologies, buildings	Kept and re-aimed: civics become economic levers; buildings become capital	The tree and the tech web stay; their effects become material

5. The ledger and the three yields

Every city keeps one ledger: for every good *g*, its stock *X*, what it made *q*, received *m*, used *u* and sent *x* this turn, in integer hundredths. Every flow in the game is a line in some ledger, so goods are conserved exactly:

$$X_{g,t+1} = \min\left(C_g,\ X_{g,t} + q_{g,t} + m_{g,t} - u_{g,t} - x_{g,t} - \left\lfloor \frac{s_g(X_{g,t} + q_{g,t} + m_{g,t} - u_{g,t} - x_{g,t})}{100} \right\rfloor\right) \quad (1)$$

The three yields become views of the ledger, so the city screen, the growth rule, the build queue and the sliders work unchanged:

$$\text{Food} = \frac{1}{100} \sum_{g \in \mathit{staple}} (q_g + m_g + \text{draw}_g) \quad (2)$$

$$\text{Production} = \frac{1}{100} L_{work}\, \pi_0\, (1+\tau)(1+\kappa) + P_{ind} \quad (3)$$

$$\text{Commerce} = \frac{1}{100}\left(\sum_g p_g \,\text{sold}_g\, \mu + \text{services} + \text{convoys}\right) \quad (4)$$

Food is the staple a city has to eat this turn, so `CvCity::foodDifference` = Food − 3 × population keeps its meaning. Production is the work of the citizens not producing goods, times their tools τ and power κ, plus P_{ind}, what industry produces as construction capacity. Commerce is what the city's market earns from goods sold at price *p* with margin μ, plus services (merchants, banks) and convoys. The sliders split commerce into gold, research, culture and espionage exactly as now. The engine's percentage modifiers (`CvCity::getBaseYieldRateModifier`) stay, applied to these views, so every building and civic that changes a yield keeps working while it is re-aimed at goods.

6. Land

Land is described in *Land, not tokens* and adopted whole: per plot, a climate (annual temperature, seasons that turn with the hemisphere, water, growing season), a soil that wears with farming and recovers, geological provinces with ore grade and three layers of reserves, and for water, a carrying capacity for fish. On the 30 games' map the reconstruction checked it against the games themselves: the rebuilt cities' worked plots feed their observed populations on 93.1 % of city-turns.

Land use (the improvement)	What it makes	Depends on
Farm (of a crop the civilization knows)	Staple or a cash crop	Crop suitability, soil, irrigation; Latifundium stages as estates
Pasture	Meat, dairy, wool, hides, horses	Grass, climate; the breed known
Orchard, vineyard, olive grove	Fruit, oil, wine	Mediterranean climate; years to bear
Plantation	Sugar, cotton, coffee, tea, tobacco, spices	Tropical climate; labour regime (section 11.4)
Mine, Shaft Mine, Modern Mine	Ores, coal, stone, salt	Province grade; depth reached; reserves left
Lumber camp, then forestry	Timber, charcoal	Forest; regrowth
Fishing fleet	Fish	Fishing ground capacity and stock
Town, then city district	Services, housing	Population; the land's commerce today

7. People and work

Every citizen has an occupation. The engine already distinguishes citizens working a plot from specialists; the Living Economy uses that distinction and fills it with meaning. RoM defines 19 specialist types today; they become the occupations below, defined in XML.

Occupation	Where	Makes
Farmer, herder, fisher, miner, forester	A worked plot	The plot's goods (section 6)

Occupation	Where	Makes
Artisan: smith, weaver, potter, miller, brewer, carpenter	A workshop slot of a building	The building's recipe, one run per artisan
Worker (from the industrial era)	A factory slot	The factory's recipe, several runs per worker with power
Merchant	A market slot	Commerce: margin on goods sold; convoy capacity
Scholar, priest, artist, official	Library, temple, theatre, court	Research, culture, contentment, administration
Labourer (no slot)	The city	Production: building and training work

The city governor assigns occupations by marginal value: each citizen goes where one more worker adds most to the city's living standard and output (section 15). In the model that measured this, a governor that knew the city's needs lowered the share of citizens short of fuel from 66.4 % to 53.4 % and of everyday food variety from 50.8 % to 43.5 % compared with one that ranked plots by yields alone.

Labour regimes are civics, and they change how work is done, not a percentage: tribal labour (everyone farms), slavery (cheap plantation and mine labour, unrest, no artisans above a size), serfdom (estates bound to land), free labour (artisans and merchants possible; wages), industrial labour (factory workers). Each sets which occupations exist, their output per worker and their effect on contentment.

8. Goods

About forty goods carry the whole game. The catalogue keeps today's resource types as names wherever they fit, so that existing art, Civilopedia text and requirements can be reused.

Era of first use	Raw goods	Made goods (inputs)
Ancient	Grain (wheat, rice, maize, potato), fish, meat and dairy, fruit, wool, hides, timber, stone, clay, copper, salt	Pottery (clay, fuel), bronze (copper, tin from trade), cloth (wool), beer and wine (grain, fruit)
Classical	Iron ore, tin, marble, olives, grapes, incense, dye	Iron tools and arms (iron, charcoal), oil (olives), fine cloth (wool, dye), coin metal (silver, gold)
Medieval	Horses, silk, spices, furs	Plate armour (iron), paper and books (timber, scholars), glass (sand, fuel), furniture (timber)
Renaissance	Sugar, coffee, tea, tobacco, cotton	Gunpowder (sulphur, saltpetre), ships' stores, refined sugar, porcelain
Industrial	Coal, lead, sulphur, rubber	Steel (iron, coal), machinery (steel), rails (steel), textiles (cotton), ammunition (lead, sulphur), chemicals
Modern	Oil, bauxite, uranium	Fuel (oil), aluminium (bauxite, power), plastics, electronics, consumer goods, fertilizer, medicine
Transhuman and future	(engineered)	Nanotubes, nanobots, personal replicators: the goods that end scarcity, as RoM's future era already names them

Rules for the catalogue. A good exists only if a choice turns on it: a different land, a different building, a different trade partner. Goods that would always travel together are one good (meat and dairy). Each era adds goods that make the previous era's goods more valuable (iron makes tools that make every farm more productive). Every made good has at least one raw good that some civilizations lack, so that trade is never optional for everyone. The catalogue size costs almost nothing to compute: 0.95 ms per turn at 300 cities and 100 goods (section 18.4).

9. Living standards

Each citizen consumes from four tiers. A tier is met in a city when every good of the tier is supplied at its level per citizen. Tiers are nested: a city aspires to comforts only once its everyday goods are met.

Tier	Goods (substitutes in each line)	If met	If short
Staple	grain · fish	Growth from the surplus	Starvation (the engine's rule)
Everyday	meat, dairy or fruit · wool or cloth · timber, charcoal or coal	Health, +1 contentment	Illness, –1 contentment, less work
Comfort	salt · wine or beer · pottery or furniture · (later) sugar, tea, coffee	+1 or +2 contentment; artisans and merchants possible	No unrest; no higher occupations
Luxury	spices, silk, fine cloth, dye, jewellery, porcelain	Culture, great people, contentment	Nothing lost: luxury is aspiration

Levels by development. A village (size 1 to 3) wants the base level; a town (4 to 8) 1.4 times it per citizen; a city (9 and more) 2.0 times. Contentment from a tier grows with the share met, with diminishing returns:

$$\eta_{tier} = H_{tier}(1 - e^{-c/k}), \qquad c = \frac{\text{supplied}}{\text{level} \times \text{population}} \tag{5}$$

The base levels were solved for, not chosen (section 17.3). With the targets that most citizens have their everyday goods (80 %) and half have comforts (50 %), the levels that meet them in the 30 games are, per citizen of a village and turn: everyday food 0.22, clothing 0.11, fuel 0.06; comfort 0.13.

10. Production and industry

A building with a recipe has slots; each artisan in a slot runs the recipe once per turn if the inputs are there. Output follows the scarcest input (fixed proportions), and tools and power raise output per worker:

$$n_k = \min\left(w_k\, r_k(1+\tau)(1+\kappa),\ \min_g \frac{\alpha_{g,k} X_g}{a_{g,k}}\right) \tag{6}$$

w is the number of workers, *r* runs per worker, α the building's share of a scarce input (exact proportional rationing). Tools are themselves a good that wears out: each worker using tools consumes a fraction of a tool per turn, so iron and fuel stay in demand for the whole game. Power (from the industrial era) is a network good produced by plants from fuel.

The industrial revolution is the moment workshops overtake land, which figure 2 shows happening already in the 30 games' first 150 turns. In the Living Economy it is the moment coal, iron and power let a worker run several recipes instead of one, which draws people from farms into factories and makes the empire depend on fed cities it does not feed itself.

11. Markets, money and the state

11.1 Markets

Each trade network has one market. After production and recipes, every surplus is offered and every unmet need bid; goods move from surplus to need inside the network automatically, limited by the network's transport capacity. The price of a good in the network follows its scarcity, within bounds so that no single turn can ruin a city:

$$p_g = p_g^0 \ \mathrm{clamp}\left(\left(\frac{D_g}{S_g}\right)^{1/\varepsilon_g},\ \tfrac{1}{5},\ 5\right) \tag{7}$$

Prices are shown to the player and used by the AI. Sharing within a network is what keeps cities fed: in the 30 games, staple kept in each city left 11.5 % of citizen-turns short; shared in the network 5.1 %; with granaries 1.2 %.

11.2 Money

Commerce is what the market earns: the margin on goods sold, plus services. The state taxes it, and the sliders split the tax into gold, research, culture and espionage as today. Coin metal (silver, gold) raises the margin a market can take; banks raise it further. Money is what buys goods across networks and from other civilizations; it is never created except by trade and taxation.

11.3 Buildings as capital

Buildings keep their roles but act on goods: a granary stores staple and keeps part of the food box after growth (Town Essentials keeps 30 % today); a market adds merchant slots and margin; a harbour adds sea transport capacity and fishing fleets; a forge is a workshop of smiths. Of today's 389 buildings, 146 give specialist slots, which become workshop slots; 26 add trade routes, which become transport capacity.

11.4 Civics as economic levers

Civic category	Today (examples)	In the Living Economy
Administration (Oral Law, Tablets, ...)	Oral Law: food –25 %, production –25 %, commerce –10 %	How much of the ledger the state sees and taxes; spoilage and waste in stores
Economy (Hunter-Gatherer, Barter, Coinage, Planned, ...)	Trade yield modifiers; Planned: trade gives +50 % food and production	How markets clear: barter (no money, swaps only), coinage (prices), planned (the state allocates)
Employment (Slavery, Caste System, Labor Allocation, ...)	Yields added to classes of improvements	The labour regime (section 7): which occupations exist and their output
Taxation (Tribute, Census, National Debt, ...)	Capital and yield modifiers	The share of commerce taxed, tariffs on foreign goods, debt
Trade policy (inside Economy today)	National Socialism forbids foreign trade	Free trade, mercantilism (tariffs, colonies), autarky (embargo)

The starting civic Oral Law shows why civics must become levers on the ledger rather than percentages: applied to food, its –25 % would leave the rebuilt cities unable to feed their observed populations on 26.9 % of city-turns instead of 5.9 % (Settlement Economy, section 5.3).

12. Trade

Inside a civilization goods move freely within a network (roads, rivers, harbours connect it); between a civilization's separate networks they move by convoy. **Between civilizations** they move by convoys along trade routes, under contracts or as spot sales.

Instrument	What it is	Engine
Spot sale	Goods sold now at the partner's market price	A deal item per good and quantity

Instrument	What it is	Engine
Contract	Quantity per turn, for a term, at a price; penalties for default; cancelled by war	Repeated TRADE_RESOURCES items at first (TradeData has no quantity field); a Python trade screen later
Seeds, breeding stock, techniques	The means to produce a good: the buyer can plant, breed or build it	A deal item like a technology
Package	Any combination with gold, technologies, treaties	The existing deal structure
Tariffs, embargo, blockade	A share of value taken at the border; no trade; ships cut a sea route	Civics; diplomacy; naval units on route plots

A trade route becomes a convoy's capacity: it carries goods worth its capacity per turn and earns the price difference. In the 30 games, all trade that could happen between players needed 0.42 units per route per turn with one route per city, so routes become a real constraint only as trade grows, which is the intent: harbours, roads and ships matter because they carry goods.

The AI trades by value. A deal is worth the sum of the marginal values (section 15) of what is received minus what is given, including the strategic value of giving a rival the means to produce. This replaces the random offers and fixed resource values that the diplomacy work of merge request !63 began to remove.

13. War

Units are built from production and goods (a legion needs iron arms; a ship of the line timber, cannon and canvas; a tank steel, fuel and ammunition). Armies in the field draw supplies each turn from the nearest connected city; a unit cut off loses strength. Battles consume ammunition from the industrial era on. Blockades and raids on provinces become ways to win: a country without steel cannot replace its tanks. An arsenal stores arms in peace; the model showed a 90-turn war emptying a 60-unit arsenal by its eleventh turn (Land, not tokens, section 9).

14. The eras

Era	The economy's question	What changes
Ancient	Can we feed ourselves and clothe ourselves?	Staple and everyday tiers; bronze needs copper and tin, rarely found together
Classical	Who controls iron and the roads?	Iron tools raise all work; coinage creates markets; roads join networks
Medieval	Can the towns grow beyond the land?	Guild workshops; mills (power from water and wind); estates and serfdom
Renaissance	Who controls the sea and the colonial goods?	Comfort goods from the tropics; banks; mercantilist trade
Industrial	Coal, iron and the factory	Workers run several recipes; power; railways move goods; cities outgrow their farms
Modern	Oil and the consumer	Oil and electricity; consumer goods raise living standards; global supply chains
Transhuman	Automation	Androids (RoM's specialist) and automated factories: work stops being people
Future	The end of scarcity	Replicators make goods from energy; the ledger's constraints relax by design

A Normal game lasts 700 turns (Marathon 1,470); the 30 measured games cover its first 150. Each era's goods should become important within its first third and remain useful after: the design target is that no good introduced in an era stops mattering before the era after next.

15. The AI

The AI uses one quantity for every economic decision: the marginal value of one unit per turn of each good to each of its networks, the change in its weighted living standards and output when that supply rises by one:

$$v_g = V(\mathbf{q} + \mathbf{e}_g) - V(\mathbf{q}), \qquad V = \sum_{cities} \left(\sum_{tier} \omega_{tier}\, \eta_{tier} + \omega_F F + \omega_P P + \omega_C C \right) \tag{8}$$

Computed by recomputing the network once per good, it costs 98 ms per turn at 300 cities and 100 goods on the measured host, against a late turn of about 20.8 s today (section 18.4). It drives: occupations and plots (the governor); what to build (recipes whose outputs are worth more than their inputs and labour); what to plant and where to prospect; what to offer, ask and accept in trade; when stores are worth filling; and which provinces and routes are worth a war. Because every decision uses the same values, the AI cannot be inconsistent across them, which is the failure mode of today's separate fixed weights.

16. The interface

- **City ledger screen** (Python): every good made, used, stored, received; each tier's share met, with the citizens short and why.
- **Land overlays**: crop suitability, ore grade and depth, fishing grounds, climate; on the map through coloured plots, the API the strategy overlay already uses.
- **Market screen**: the network's prices and their history; what is short; what sells.
- **Contracts and convoys**: what flows in and out, with whom, until when.
- **Plot and city help**: every number names its cause (this plot, this worker, this building, this civic).
- **Advisors**: shortages and opportunities in one line each: "Iron town short of grain for 3 turns: the road to the farms is cut".

17. Numbers and where they come from

17.1 Anchors from the engine

Quantity	Value	Source
Staple per citizen	3 per turn	FOOD_CONSUMPTION_PER_POPULATION (GlobalDefines.xml)
Growth threshold	30 + 10 × size	BASE_CITY_GROWTH_THRESHOLD, CITY_GROWTH_MULTIPLIER; CvPlayer::getGrowthThreshold
Plots per city	21	NUM_CITY_PLOTS
Game length	700 turns Normal, 1,470 Marathon	CIV4GameSpeedInfo.xml
Yields and commerce outputs shown	3 and 4	YieldTypes and CommerceTypes, compiled into the executable
Deal item	a type and one integer	TradeData (CvStructs.h), exported to the executable
Mine depths	Mine, Shaft Mine, Modern Mine: production +1, +2, +8	CIV4ImprovementInfos.xml
Specialist types today	19	CIV4SpecialistInfos.xml

17.2 Measured in the 30 games

Quantity	Value	Source
Rebuilt cities fed by their plots	93.1 % of city-turns	Settlement Economy, section 4
Production over plots, turn 150	4.08 (median player)	Settlement Economy, section 5.3
Trade routes at the 1.00 floor, turn 150	2708 of 3025	Settlement Economy, section 5.4
Staple short: local / shared / with stores	11.5 % / 5.1 % / 1.2 %	Settlement Economy, section 7.1
Production lost to whole units	8.8 %	Settlement Economy, section 7.5
Allocations changed by floating-point order	1803 of 120323	Settlement Economy, section 7.5
Resource deals between AIs that paid gold	1 of 580	Settlement Economy, section 5.1
Cost per turn, 300 cities and 100 goods	0.95 ms; 98 ms with marginal values	Settlement Economy, section 7.8

17.3 Need levels solved from targets

The targets are design intents: everyday goods for 80 % of citizens, comforts for 50 %, and staple shortage no worse than today's rules (9.2 %). Under the recommended rules (staple shared, granaries, spoilage, a governor that knows needs; levels by development, a town 1.4 and a city 2.0 times a village), a bisection over one level per need, run on all 30 games each step, gives the levels in section 9. With them, 81.1 % of citizens have everyday food, 80.0 % clothing, 80.3 % fuel and 49.8 % comforts, and 5.5 % of citizen-turns are short of staple.

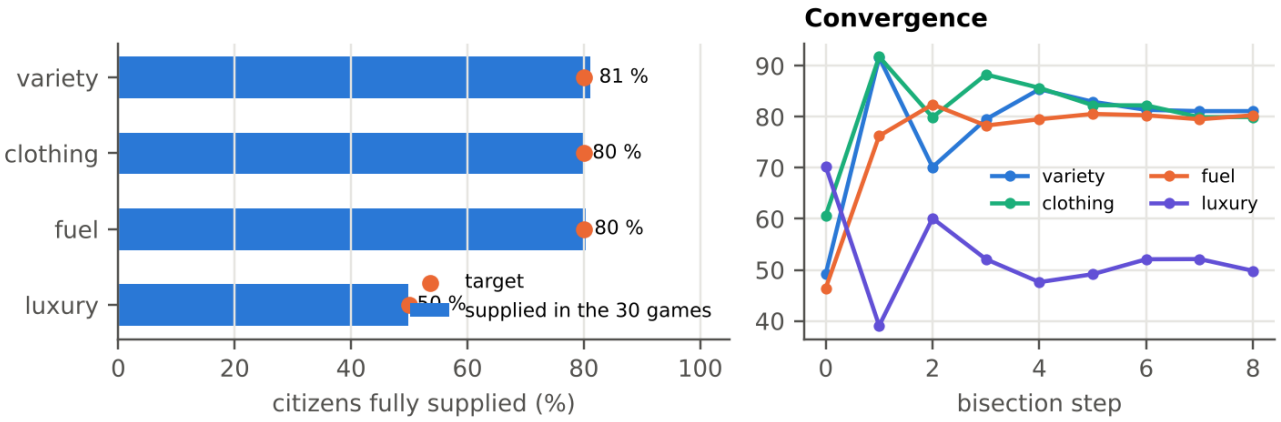


Figure 3. Left: the share of citizens fully supplied with each need at the solved levels, against its target. Right: how the bisection converged.

These levels hold for the land alone in the first 150 turns of one scenario. Each later era adds workshops and imports to the supply and must be solved the same way, with the scenario set widened to the map types RoM ships.

18. Architecture

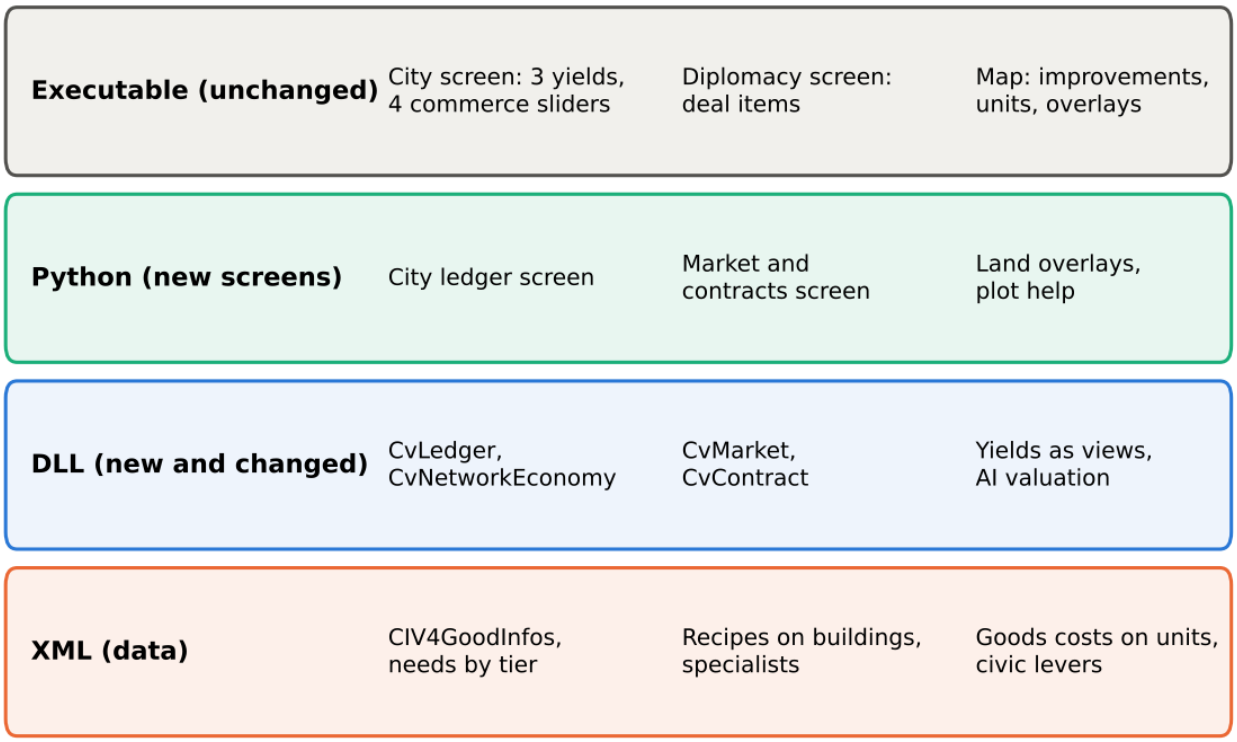


Figure 4. Where each part lives. The executable is not changed; everything new is DLL, XML and Python.

18.1 New classes

Class	Holds	Saved
CvGoodInfo (XML: CIV4GoodInfos.xml)	Unit, tier, spoilage, storage, base price, elasticity, the resource type it reuses for art and text	no
CvLedger (in CvCity)	Per good: stock, made, used, received, sent (int hundredths)	yes, behind CvCity's uiFlag
CvNetworkEconomy (per CvPlotGroup)	Recipes in dependency order, rationing, transport capacity, the market	derived each turn
CvMarket	Prices and their history per good	yes
CvContract (per player pair, in CvDeal)	Good, quantity per turn, term, price, penalty	yes

18.2 Changed functions

- `CvPlayer::doTurn`: the economy pass (figure 5) before the loop over cities.
- `CvCity::getBaseYieldRate`: food, production and commerce as views of the ledger (equations 2 to 4).
- `CvCity::foodDifference`, `doGrowth`: unchanged in form; they read the views.
- `CvPlot::calculateYield` and improvement yields: from land qualities, not resources.
- `CvCity::changeFreeBonus`: the bridge that lets resource-type requirements read the ledger during the transition.
- `CvDeal`: contracts; `CvPlayerAI`, `CvCityAI`: the valuation engine.

- Map scripts and WorldBuilder loading: provinces from the old resource plots, then the plots cleared.

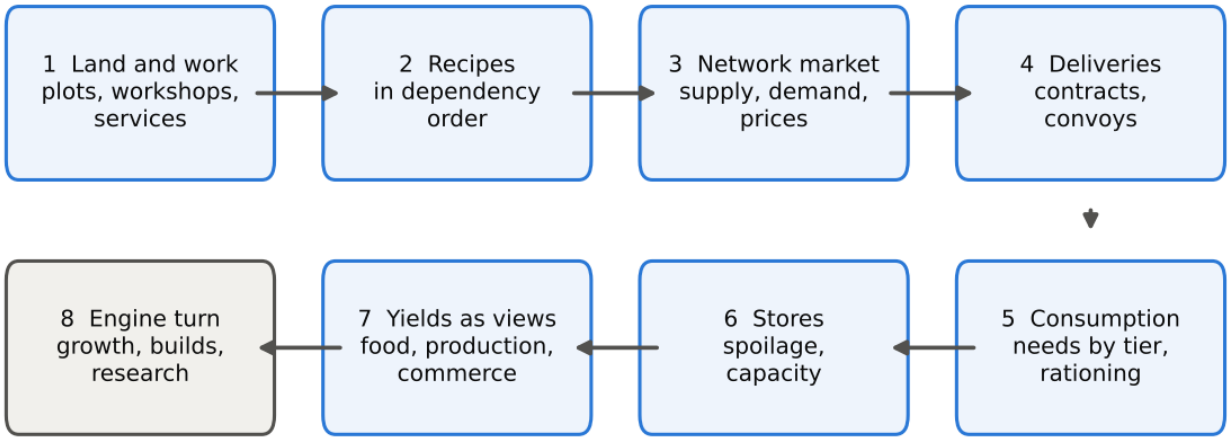


Figure 5. One economy turn, per trade network, before the engine's own turn.

18.3 Determinism and saves

All rules are integer arithmetic in hundredths, rationing is exact (largest remainders, ties by index), and every random draw (harvests) uses the synchronised generator. New data sits behind the version flags the classes already read first. Old saves load with empty ledgers.

18.4 Performance budget

Measured in C on the study host: 0.95 ms per turn for the goods pass at 300 cities and 100 goods, 98 ms with the AI's marginal values; at 600 cities and 40 goods, 0.91 ms and 37 ms. The budget for the whole economy is 1 % of a turn; at today's late-turn time of about 20.8 s that is 200 ms.

19. Building it

The Living Economy is built on one branch, as a whole, not merged piece by piece into today's economy. The branch is playable at every milestone because the bridge (CvCity::changeFreeBonus) keeps every unconverted rule reading goods as resources.

Milestone	Content	Done when
M1 Ledger and land	Goods XML; CvLedger; land qualities; plots make goods; staple as a shared good; the three yields as views; the bridge	Harness: no crash or slower turns (5 seeds); food shortage within 1 point of today's; growth curves within 10 %
M2 People and living standards	Occupations; workshops; tiers and contentment; the needs-aware governor; the ledger screen	Harness: tier shares in the target bands of section 17.3 (30 seeds)
M3 Markets and money	Network markets, prices, commerce from sales; civics as levers	Harness: commerce per citizen within 15 % of today's by era
M4 Trade	Convoys, contracts, seeds, the AI's deal valuation; the Python trade screen	Harness: AI trade volume and its balance; the user's games
M5 Industry and war	Recipes with tools and power; units costing goods; supply; ammunition	Harness: war outcomes against fixed opponents (30 seeds)
M6 Later eras and balance	Industrial, modern and future goods; need levels per era solved again	The user's games through the late game

Each milestone is a registered prediction judged under the repository's rules. The deterministic model of Settlement Economy is extended with each milestone first, so every rule is measured on real cities before it is written in the DLL.

20. Risks

Risk	How the design answers it
The AI cannot play it	One valuation engine for every decision; measured affordable; each milestone judged on AI outcomes first
Too much to manage	Distribution is automatic; governors assign work; contracts run themselves; the player sets direction
Balance collapses when one part changes	Need levels and prices are solved from targets, not tuned by hand; the model re-solves them for every change
Multiplayer goes out of sync	Integer arithmetic and exact rationing; floating point was shown to change allocations
The executable's screens cannot show it	Three yields and four commerce outputs kept as views; new information in Python screens
Old saves and scenarios	Provinces from old resource plots; version flags; empty ledgers on load
Later eras untested	M6 re-solves them; the 30 games cover only the first 150 of 700 turns